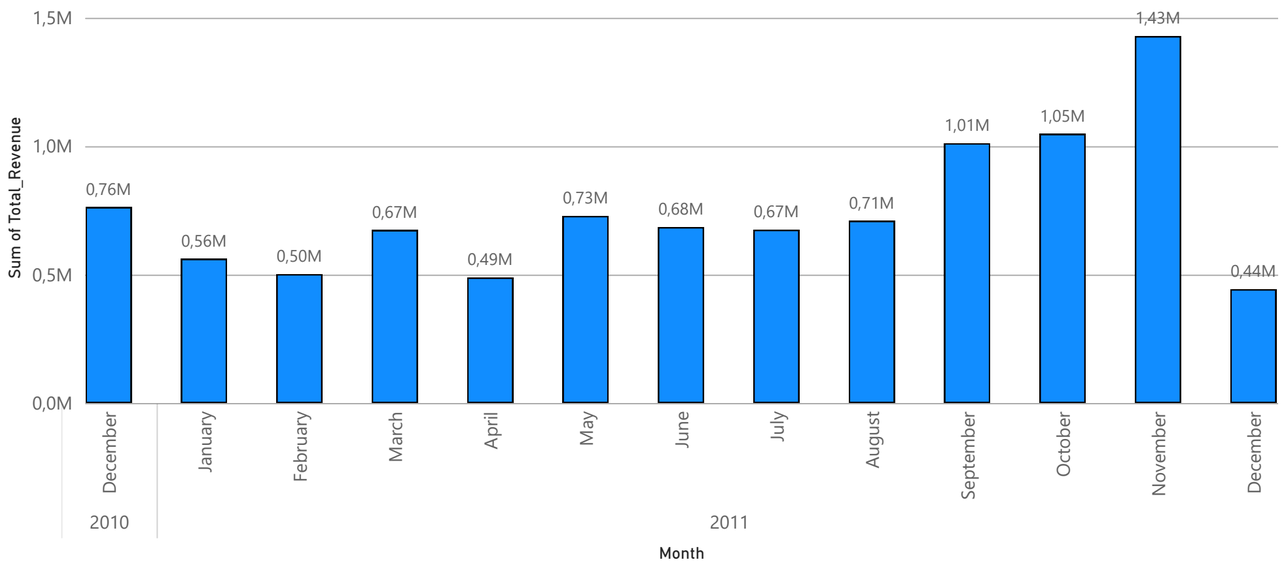
BUSINESS OVERVIEW

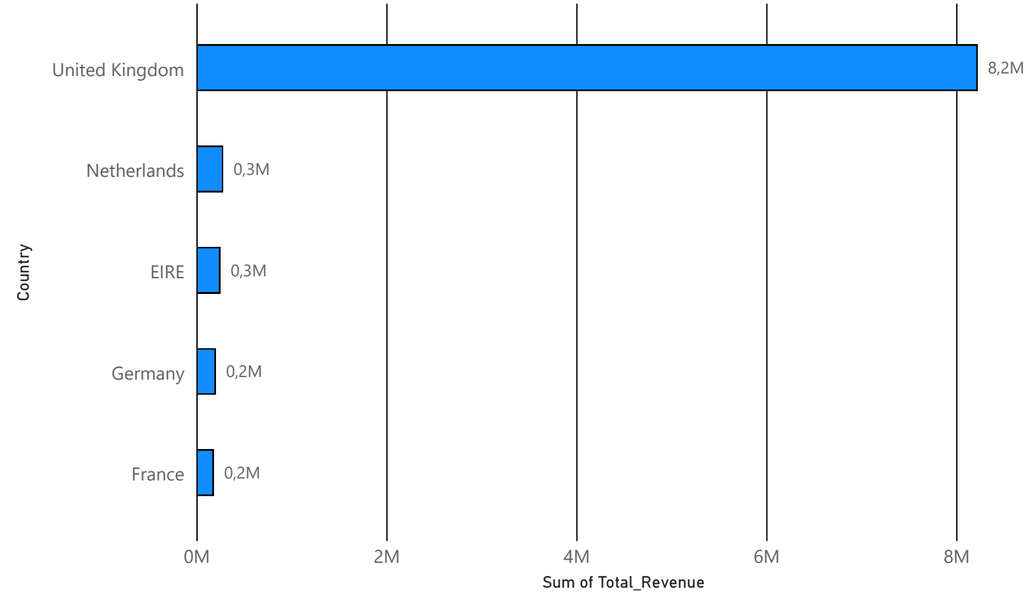
Total Revenue	Peak Month Revenue	UK Share of Top 5
9,71M	1,43M	90,0%

Revenue peaked in November 2011 ahead of Christmas

Sum of Total_Revenue by Year and Month



UK is about 90% of revenue among top 5 markets

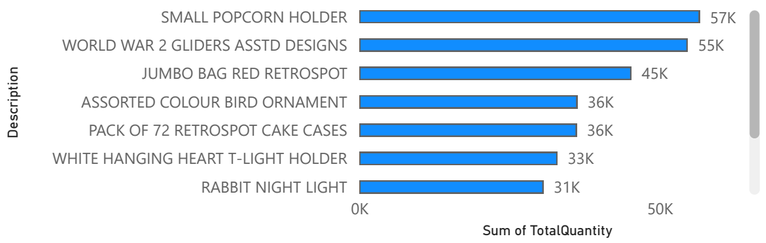


- **Sales built up strongly from October to November.**
- **December 2011 is partial : the data stops on 9 December.**

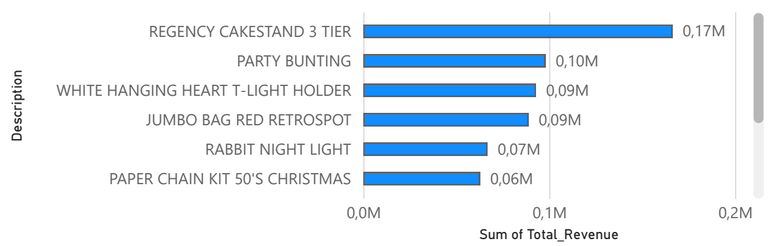
PRODUCT PERFORMANCE

Top 10 Units	Top 10 Revenue	Top 10 Share of Revenue
370K	430,02K	4,4%

Top 10 Products by Units Sold



Top 10 Products by Revenue



High volume does not always mean high revenue



Products that sold only one unit

Description	Units sold
*Boombox Ipod Classic	1
CAPIZ CHANDELIER	1
COUNTRY COTTAGE DOORSTOP GREEN	1
Dotcomgiftshop Gift Voucher £100.00	1
ETCHED GLASS STAR TREE DECORATION	1
GOLD COSMETICS BAG WITH BUTTERFLY	1
KEEP OUT GIRLS DOOR HANGER	1
MUMMY MOUSE RED GINGHAM RIBBON	1
SCALLOP SHELL SOAP DISH	1
SET OF THREE VINTAGE GIFT WRAPS	1
Total	**10**

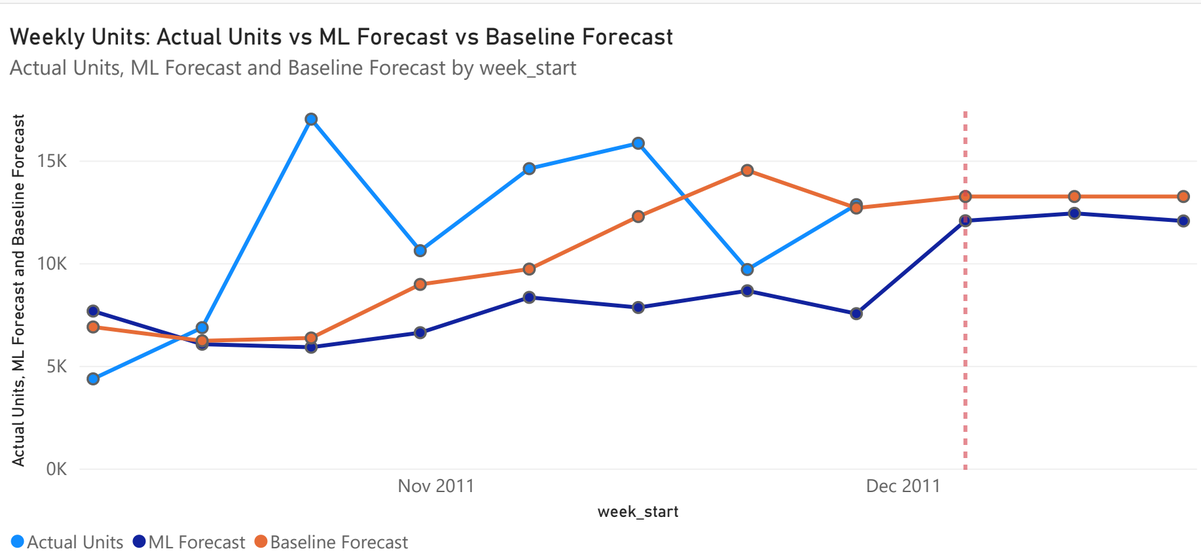
Choosing the top 10 by quantity focused the forecast on fast movers. Revenue leaders such as the Regency Cakestand are a natural next step.

DEMAND FORECAST: NEXT 3 WEEKS

Top 10 Products

All

Baseline Next 3 Weeks	ML Next 3 Weeks
39,74K	36,55K



Description	Baseline Next 3 Weeks	ML Next 3 Weeks
WORLD WAR 2 GLIDERS ASSTD DESIGNS	2 955,60	2 337,80
WHITE HANGING HEART T-LIGHT HOLDER	2 639,40	2 167,50
SMALL POPCORN HOLDER	9 801,00	9 490,70
RABBIT NIGHT LIGHT	10 248,60	9 255,50
PACK OF 72 RETROSPOT CAKE CASES	1 377,60	1 820,60
PACK OF 60 PINK PAISLEY CAKE CASES	975,00	1 802,80
PACK OF 12 LONDON TISSUES	1 754,40	2 173,70
MINI PAINT SET VINTAGE	2 329,50	1 923,60
JUMBO BAG RED RETROSPOT	3 895,50	2 507,90
ASSORTED COLOUR BIRD ORNAMENT	3 768,00	3 074,20
Total	**39 744,60**	**36 554,30**

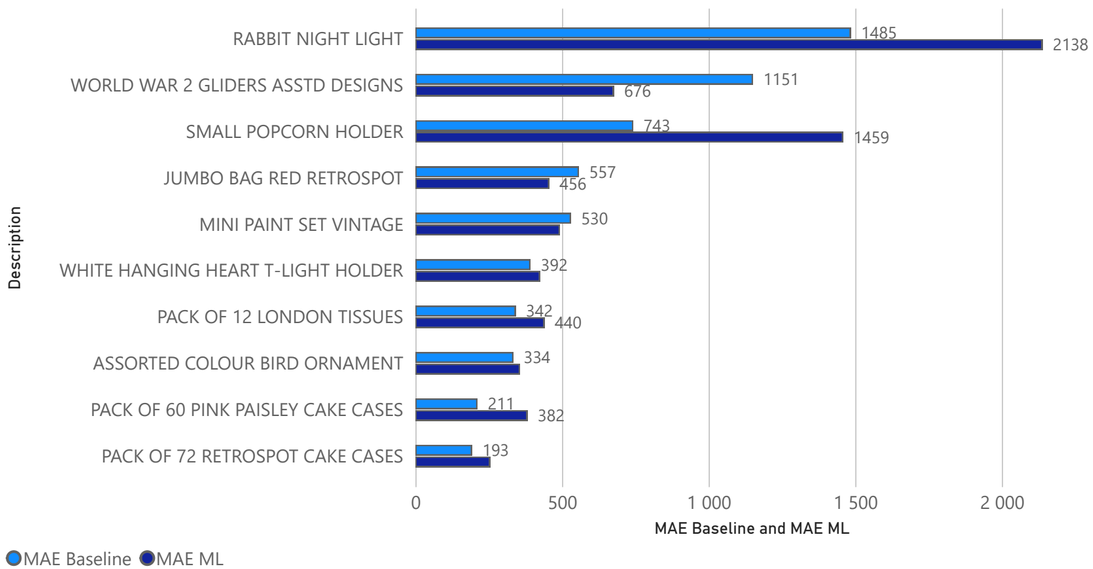
ML forecast after the red line comes from the model retrained on all weeks, so it has not been tested. Week of 26 Dec not shown: shop assumed closed.

MODEL EVALUATION

MAE Baseline	MAE ML	RMSE Baseline	RMSE ML
593,87	708,12	1,03K	1,20K

The baseline had the lower average error overall (MAE 594 against 708 units). The ML model struggled most on the two products with sudden jumps in demand, Rabbit Night Light and Small Popcorn Holder: their actual sales rose to 2,000 to 5,000 units a week while the model stayed near 1,000 to 1,300. The ML model did better on World War 2 Gliders, Jumbo Bag Red Retrospot and Mini Paint Set Vintage.

The ML model beat the baseline on 3 of 10 products



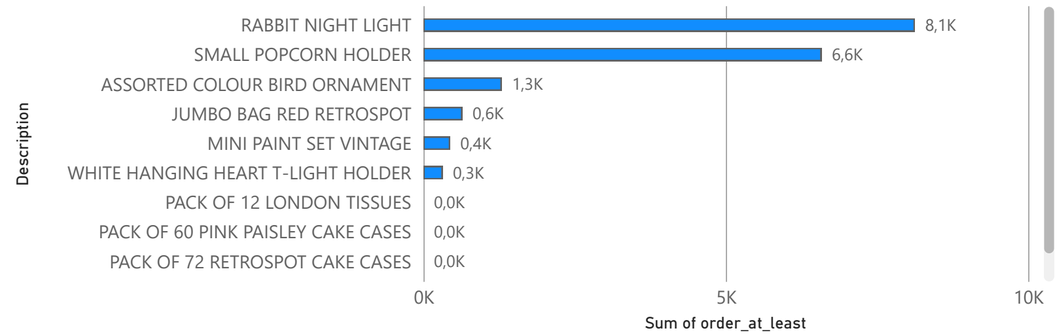
Description	MAE Baseline	MAE ML	Better Model
ASSORTED COLOUR BIRD ORNAMENT	334,13	356,71	Baseline
PACK OF 12 LONDON TISSUES	341,84	439,96	Baseline
PACK OF 60 PINK PAISLEY CAKE CASES	211,46	381,98	Baseline
PACK OF 72 RETROSPOT CAKE CASES	192,74	254,78	Baseline
RABBIT NIGHT LIGHT	1 484,59	2 138,38	Baseline
SMALL POPCORN HOLDER	742,88	1 458,81	Baseline
WHITE HANGING HEART T-LIGHT HOLDER	392,18	425,10	Baseline
JUMBO BAG RED RETROSPOT	557,48	456,31	ML model
MINI PAINT SET VINTAGE	530,35	492,70	ML model
WORLD WAR 2 GLIDERS ASSTD DESIGNS	1 151,08	676,43	ML model
Total	**593,87**	**708,12**	**Baseline**

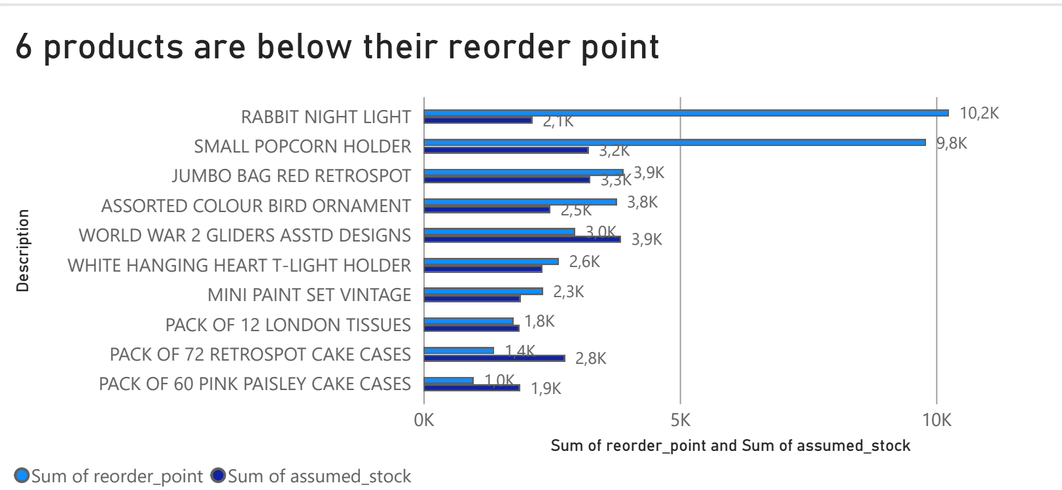
STOCK RECOMMENDATION

Products To Reorder
6

Units To Order
17K

Description	Sum of weekly_forecast	Sum of assumed_stock	Sum of reorder_point	Sum of order_at_least	reorder
RABBIT NIGHT LIGHT	3416	2124	10249	8125	Yes
SMALL POPCORN HOLDER	3267	3220	9801	6581	Yes
JUMBO BAG RED RETROSPOT	1298	3255	3896	641	Yes
ASSORTED COLOUR BIRD ORNAMENT	1256	2475	3768	1293	Yes
WORLD WAR 2 GLIDERS ASSTD DESIGNS	985	3853	2956	0	No
WHITE HANGING HEART T-LIGHT HOLDER	880	2323	2639	316	Yes
MINI PAINT SET VINTAGE	776	1891	2330	439	Yes
PACK OF 12 LONDON TISSUES	585	1873	1754	0	No
PACK OF 72 RETROSPOT CAKE CASES	459	2760	1378	0	No
PACK OF 60 PINK PAISLEY CAKE CASES	325	1881	975	0	No
Total	**13247**	**25655**	**39746**	**17395**	





ASSUMPTIONS

Stock levels are assumed because the dataset has none; weekly forecast is the baseline average for 5 to 19 December; reorder point is 3 weeks of forecast demand; order quantity is the reorder point minus assumed stock

The forecast says what will sell; this page says what to do about it. Rabbit Night Light and Small Popcorn Holder need urgent attention; four products are sufficiently stocked.